System Analysis and Design

Group Project — Assignment 2

Part II: Design | Weight: 30% | 2026-2027, 1st Semester

For Part II of the project, you should turn your Part I requirements into a system design, covering the logical, deployment, and technology architecture, supported by visual design models (such as C4, UML, and SysML diagrams) and the key design decisions with their rationale. As in real practice, you may refine your requirements as the design exposes gaps or conflicts; note what changed and why. If your system is agentic, the design should also address its special concerns, such as agent roles, tool interfaces, memory and state, orchestration, and guardrails.

The quality of your design decisions matters more than the volume of your documentation. You are encouraged to use other graphical or visual representations that convey your design more efficiently. Present your progress in class (Week 9 or Week 10) and then submit your final design artifacts by the end of Week 10, incorporating the feedback you received.

1. Purpose and Design Mindset

Approach this assignment as a software designer and architect, not simply as someone translating requirements into components. Adopt a problem driven mindset: explore the business domain to understand real needs and constraints, and investigate the technology landscape to understand capabilities, limitations, and trade-offs. Good design begins with understanding the problem, not choosing a solution. Think beyond individual components to the system as a whole, evaluate alternatives against the system's goals, and justify not only what you choose, but why.

Your design should provide an architectural blueprint for implementation, supported by detailed models of critical parts and explicit rationale for key design decisions. Make the overall structure coherent while examining complex or consequential parts in greater depth. In the next assignment, Part III, you will use agentic development to build prototypes and proofs of concept (PoCs), test your design decisions, uncover architectural limitations, and iteratively refine your design. Aim for a design that is concrete enough to guide implementation and explicit about its assumptions, trade-offs, and potential weaknesses.

2. Presentation and Submission

Requirements and Change Management	Start from your Part I SRS and the feedback you received. As in real practice, refine requirements when the design exposes gaps or conflicts, and record what changed and why.
Presentation	In class, Week 9 or Week 10; your team's session will be announced on Canvas. **6 min per team**, all members present
Slides	Upload to Canvas before your session starts
Final submission	Canvas, by **2026/11/20, 19:00** (end of Week 10), with all the design artifacts (see Section 3)

3. Design Artifacts

Submit your design artifacts on Canvas, including a short design summary, your models and diagrams, and the supporting records listed below. Use appropriate visual and formal representations to communicate the system structure, specify key design decisions, and support shared understanding within your team. Choose formats that best convey each artifact's purpose and level of detail. Keep the documentation concise, focusing on clarity, rationale, and design quality. Ensure your design is sufficiently clear to guide implementation and support ongoing discussion, review, and continuous refinement.

No.	Artifact	Description
1	Design Summary (~ 500 words)	Progress since Part I; what changed and why; current status.
2	Architecture and Key Decisions	Logical, deployment, and technology architecture, with decision rationale.
3	Design Models	Domain, structural, and interaction models that communicate system structure and behavior.
4	Updated Requirements	Refinements to your Part I requirements, with a change log explaining what changed and why.
5	UI and UX Design	Interface mockups and user flows illustrating key interactions and user experience.
6	Open Issues and Risks	Unresolved questions, design uncertainties, and decisions to evaluate in the Part III prototype or proof of concept.
7	AI Tools Reflection	Use of AI tools, their contributions and limitations, and critical assessment of their outputs.
8	Annotated References (200-300 words each)	Relevant sources and their influence on design choices.
9	Team member contributions	Each member's contributions to analysis, modeling, design, and decision making.

A. Notes on Architecture and key decisions

Demonstrate systems-level thinking across three architectural views:

- **Logical architecture**: Major subsystems or components, responsibilities, and relationships.
- **Deployment architecture**: Runtime environments, deployment nodes, components distribution.
- **Technology architecture**: Selected languages, frameworks, platforms, services, and models, including alternatives considered
- **Key decisions**: Alternatives and rationale, documented in a decision log or ADRs (Architecture Decision Records).
- **Quality attributes:** Priorities such as performance, security, and maintainability, and how the architecture addresses them.

- **Architectural risks:** Key risks and mitigation strategies.
- **System level diagram:** At least one architecture diagram (C4, UML, SysML, or another visual form), accompanied by a narrative.
- **Agentic systems:** Address relevant agent specific design concerns.

An agile solution architect does not decide everything upfront. Identify the decisions that matter most, justify them carefully, and leave room for less critical choices to evolve.

B. Notes on Design models

Understand the problem before committing to solutions. Model use case realizations where appropriate:

- **Class diagrams:** Key abstractions, responsibilities, and important attributes.
- **Interaction diagrams:** Sequence or collaboration diagrams showing how components work together to realize important use cases.
- **Notation:** C4, UML, SysML, or any graphical or visual representation that communicates and specifies the design effectively.

4. Encouraged Practices

The following practices are encouraged:

- **Reflection:** Revisit Part I assumptions and requirements, explaining what changed and why.
- **Continuous architecting:** Evolve the architecture as your understanding develops.
- **Design thinking:** Reframe the problem when underlying assumptions prove questionable.
- **Design for validation:** Identify critical decisions that a prototype or PoC can test.
- **Agentic patterns:** Where applicable, consider planner and executor separation, tool use boundaries, human oversight, and appropriate levels of agent autonomy.